



Software Requirements Specification

For
DocuLaTeX

Version 1.4

24 August 2025

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1. Introduction

1.1 Purpose

This document defines the functional and non-functional requirements for the DocuLaTeX application. Its primary purpose is to provide a clear and unambiguous specification for a web-based tool that converts documents from common formats (PDF, Markdown, HTML) into a themed LaTeX output.

1.2 Document Conventions

This document follows the IEEE 830-1998 standard. All requirements are uniquely identified for traceability. British International English spelling and grammar are used throughout.

1.3 Intended Audience and Reading Suggestions

This document is intended for project managers, web developers, UI/UX designers, the quality assurance team, and project stakeholders involved in the development of DocuLaTeX.

1.4 Product Scope

Product Scope

DocuLaTeX is a web-based service designed to automate the process of formatting documents into a high-quality, consistent, and professional layout using a predefined LaTeX theme. It accepts user-uploaded PDF, Markdown, and HTML files and generates a downloadable PDF and the source `.tex` file as output.

Version 1.0 (MVP) Scope: The initial release will focus on the core conversion engine delivered through a simple web interface. The scope includes: a user account system, a file upload mechanism, a server-side parsing module for each supported format, a mapping engine to convert parsed elements to the specified LaTeX theme, and the generation of the final PDF and `.tex` files for download.

1.5 References

-  IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*.
-  AUCDT Brand Identity Guide v2.0, August 2025.

- ☐ The LaTeX source of *srs.tex* for CounselorPro, which serves as the target theme definition.

2. Overall Description

2.1 Product Perspective

DocuLaTeX is a new, self-contained, standalone web application. It will be accessible via a web browser and will not require any local software installation for the end-user. The server-side component will manage all processing, including LaTeX compilation.

2.2 Product Functions

The major functions are: User Account Management, File Upload, Content Parsing, LaTeX Generation and Compilation, and Output File Delivery (Preview and Download).

2.3 User Classes and Characteristics

User Classes

Academics & Researchers: Need to quickly format papers and articles into a consistent style without installing or managing a LaTeX environment.

Technical Writers: Require a tool to automate the conversion of documentation from Markdown or HTML into print-ready PDFs.

Students: Want an easy, accessible way to make their reports and assignments look professional.

2.4 Operating Environment

The application will be a web-based platform hosted on a secure cloud infrastructure. It must be accessible through modern web browsers (latest versions of Chrome, Firefox, Safari, Edge) on desktop devices. The server environment must include a full LuaLaTeX distribution for document compilation.

2.5 Design and Implementation Constraints

❗ Important Note

The primary constraint is the fidelity of the conversion. The generated PDF must be a high-fidelity representation of the source content, styled according to the target theme. The server must securely handle user-uploaded files.

- ⚠ C-1 (Server-Side LaTeX):** The server environment is responsible for all LaTeX compilation. This requires robust sandboxing to securely process user-provided content.
- ⚠ C-2 (Parsing Limitations):** The parsing engines must be robust but are not expected to handle poorly-formed or exceptionally complex source files in V1.0. For PDFs, only text-based documents are supported.
- ⚠ C-3 (Theme Hardcoding):** The target LaTeX theme (based on the AUCDT style) is hardcoded for V1.0.
- ⚠ C-4 (Browser Compatibility):** The user interface must be functional and visually consistent on the latest two versions of major desktop web browsers.

2.6 Assumptions and Dependencies

Assumption: Users have a stable internet connection for uploading and downloading files.

Assumption: Input files are well-structured. For example, HTML uses semantic tags, and PDFs have selectable text.

Dependency: The system depends on third-party cloud hosting and server-side libraries for parsing and file management.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces (UI)

The primary interface will be a graphical web interface, designed to be clean, intuitive, and conformant to the AUCDT brand identity.

The web interface shall consist of the following key components:

-  **Dashboard:** A main page displaying user's recent documents and an option to start a new conversion.
-  **File Upload Zone:** A clearly marked area where users can drag-and-drop or browse to select a single file (.pdf, .md, .html) for conversion.
-  **Processing View:** An interface that shows the status of the conversion (e.g., "Uploading," "Parsing," "Compiling LaTeX," "Done").
-  **Preview & Download Panel:** Once processing is complete, this panel will show a preview of the generated PDF and provide clearly labelled buttons to download the final PDF and the source .tex file.

3.2 Functional Requirements

ID	Requirement Description
FR-1: User Account Management	
FR-1.1	The system shall allow users to register for a new account using an email and password.
FR-1.2	The system shall allow registered users to log in and log out.
FR-1.3	The system shall associate uploaded documents and their outputs with the logged-in user's account.
FR-2: File Upload and Validation	
FR-2.1	The system shall provide an interface for an authenticated user to upload a single file.
FR-2.2	The system shall validate that the uploaded file has a supported extension (.pdf, .md, .html) and is below a maximum size of 10 MB.
FR-3: Content Processing	
FR-3.1	The system shall extract structural elements from the uploaded file (headers, lists, tables, etc.).
FR-3.2	The system shall map parsed elements to corresponding LaTeX commands based on the predefined theme.
FR-3.3	The system shall compile the generated .tex file on the server to produce a PDF.
FR-4: Output Delivery	
FR-4.1	The system shall display an embedded preview of the first page of the generated PDF.

ID	Requirement Description
FR-4.2	The system shall provide a button for the user to download the full generated PDF file.
FR-4.3	The system shall provide a button for the user to download the generated source <code>.tex</code> file.

3.3 Non-Functional Requirements

3.3.1 Performance

NFR-1.1: Web pages shall load in under 3 seconds on a standard broadband connection.

NFR-1.2: The end-to-end conversion process for a 10-page document shall be completed within 30 seconds.

NFR-1.3: The system shall support up to 20 concurrent conversion processes.

3.3.2 Security

NFR-2.1: All user passwords must be hashed using a strong, modern algorithm (e.g., Argon2).

NFR-2.2: User-uploaded content must be handled in a secure, sandboxed environment to prevent execution of malicious code on the server.

3.3.3 Software Quality Attributes

★ **NFR-3.1 (Correctness):** The generated PDF must be a high-fidelity, visually correct representation of the source document formatted with the theme.

★ **NFR-3.2 (Usability):** The web interface must be intuitive, requiring no training for a user to upload a file and get a result.

3.4 Error Handling and Recovery

❗ **EH-1:** If a user uploads an unsupported file type or a file that exceeds the size limit, the UI shall display a clear, user-friendly error message in real-time.

❗ **EH-2:** If the server-side compilation fails, the system shall inform the user with a message like, "We couldn't convert your document. Please check if the file is corrupted or badly formatted and try again."

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